Materials for Energy Devices(MDSE)

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Highly Sensitive Flexible Footstep Energy Harvester Sensor Using PZT Thin Film on PVDF Substrate

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ABSTRACT :

A Highly flexible piezoelectric energy harvester is presented, featuring a hybrid arrangement of PZT thin film and PVDF supported by a silicone substrate. The PZT layer delivers high electrical output, while the PVDF provides mechanical softness and allows the device to conform to curved or deforming surfaces. Flexibility and long-term stability under concentrated pressure are achieved by using PVDF to distribute strain and by adopting an interdigitated PZT layout that minimizes local stress. The silicone base further supports strain transfer and impact absorption. This hybrid design is aimed at integration into foot insoles and seating surfaces, enabling pressure-tolerant, self-powered sensing in everyday environments.

Key words:
Piezoelectric energy harvesting, PZT thin film, PVDF substrate, interdigitated structure, flexible sensing, strain distribution, pressure-tolerant design, self-powered systems.

I. Introduction

Flexible piezoelectric energy harvesters are increasingly explored for powering wearable devices and monitoring human motion, with particular interest in utilizing the repetitive pressure generated by human footsteps. Rigid PZT offers very high piezoelectric coefficients but is too stiff and brittle for direct implantation in the plantar region, which undergoes large, cyclic deformation (Ren et al.). In contrast, PVDF is lightweight, flexible, and able to conform to the curved geometry of the foot and insole, although its lower piezoelectric response restricts the achievable voltage and power under realistic gait conditions (Chamankar et al.). Recent studies on flexible PZT thin films and PZT/PVDF hybrid structures indicate that combining a high-performance ceramic with a compliant polymer can enhance harvested energy while maintaining flexibility, but detailed correlation between local plantar pressure and voltage output in such hybrids remains limited (Ren et al.; Chen et al.). In this work, a PZT/PVDF hybrid is implemented on a silicone substrate as an array of PVDF elements embedded within a PZT matrix, so that the PVDF layer absorbs part of the mechanical load, preserves overall flexibility, and reduces direct stress on the PZT phase, thereby mitigating fracture risk while still enabling efficient energy conversion rate.

II. Materials and Methods:

In hybrid piezoelectric architectures designed for biomechanical energy harvesting, polyvinylidene fluoride (PVDF) functions as the compliant, high-flexibility electromechanical transducer, while lead zirconate titanate (PZT) contributes a high piezoelectric charge coefficient and superior dielectric polarization stability. When these layers are co-laminated onto a soft silicone elastomer substrate, the composite structure achieves an optimized balance between mechanical cushioning, strain transfer efficiency, and durability under cyclic loading.

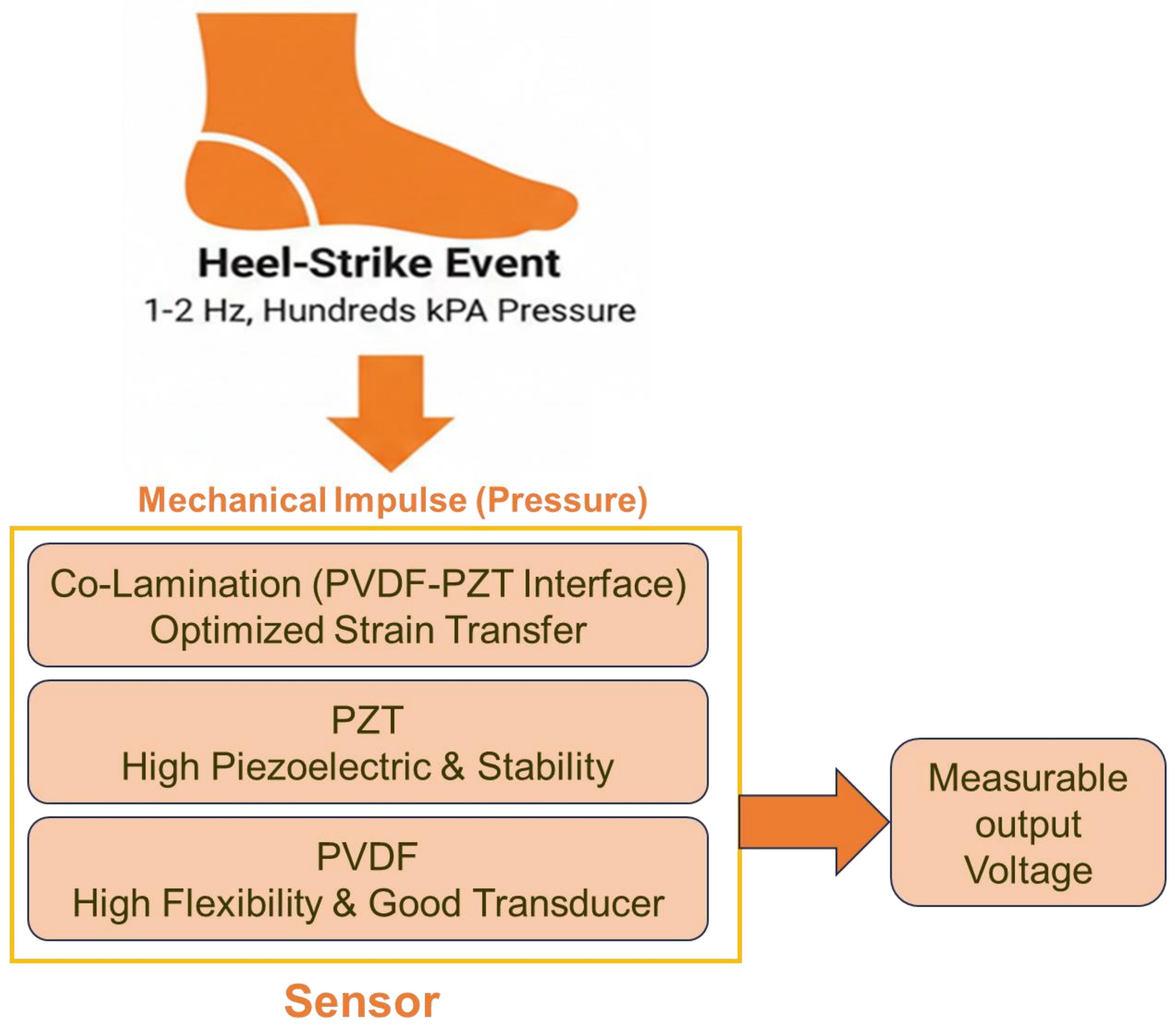

Fig.1. High Sensitive Sensor interface with PVDF and PZT.

III. Simulation & Analysis:

The piezoelectric voltage output can be modelled as:

$$V = g_{33} \cdot t \cdot \sigma \qquad (1)$$

Where g_{33} : piezoelectric voltage constant, t: active thickness σ: applied mechanical stress.

In this simulations, the PZT sections produce a larger voltage than the PVDF sections for the same applied load, due to their higher g_{33} value and greater stiffness. The interdigitated PVDF network takes up part of the deformation and allows fine bending and local strain, which increases the overall flexibility of the device and helps protect the PZT from cracking.

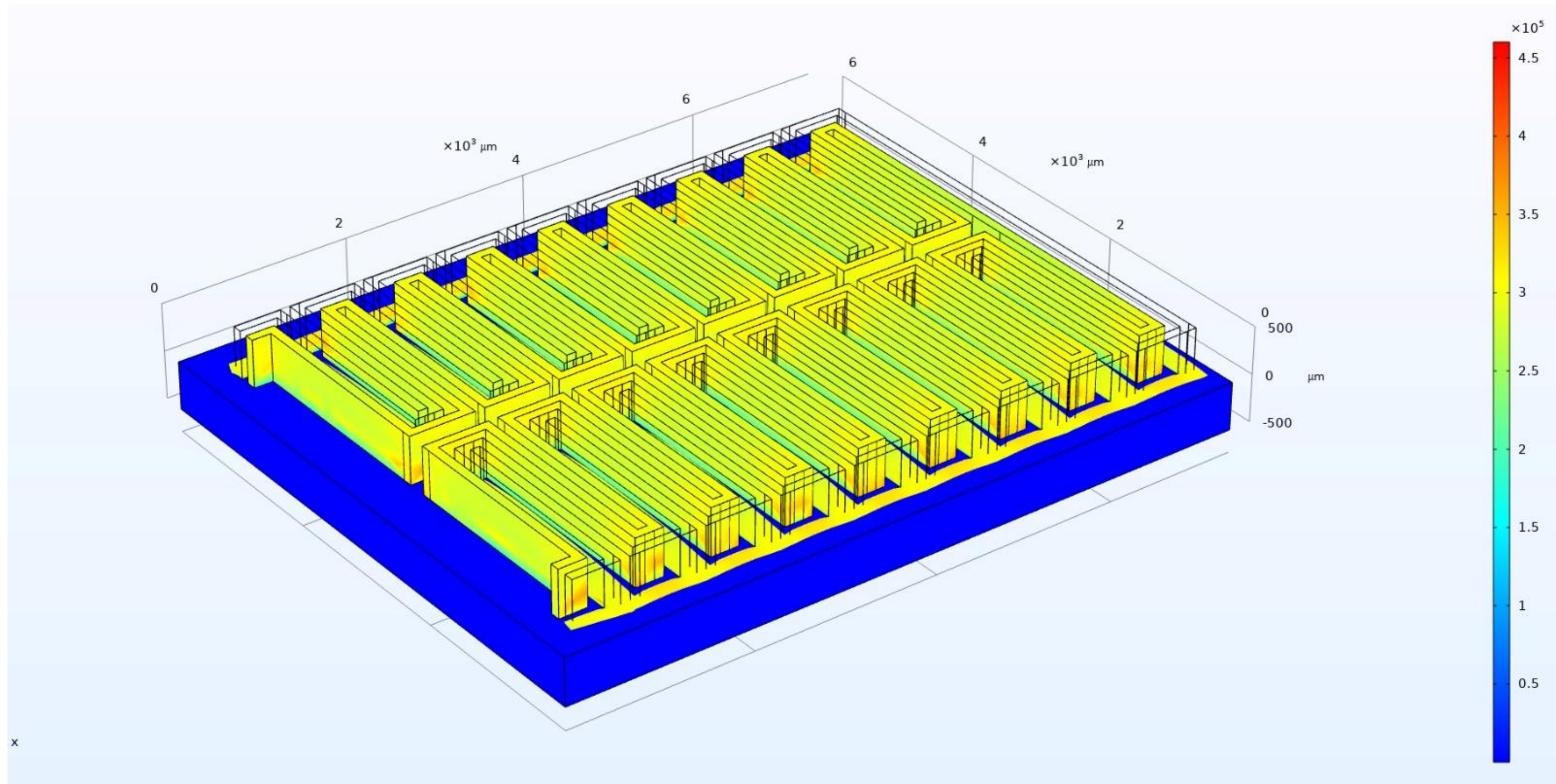

Fig.2. Simulation result of the Deformation of PVDF and PZT under pressure ranging 10kPa to 300kPa

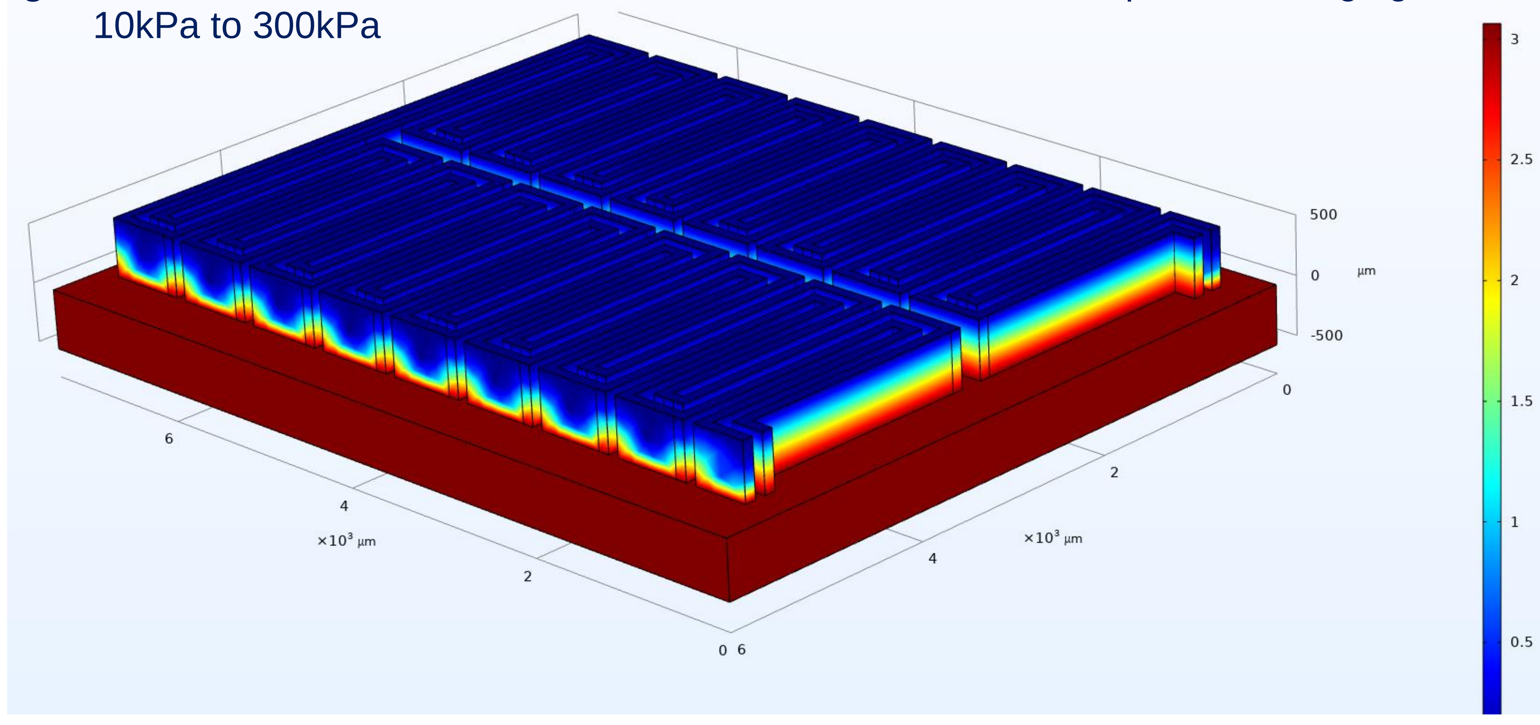

Fig.3. Simulation result of the Piezo electric phenomenon on PVDF and PZT

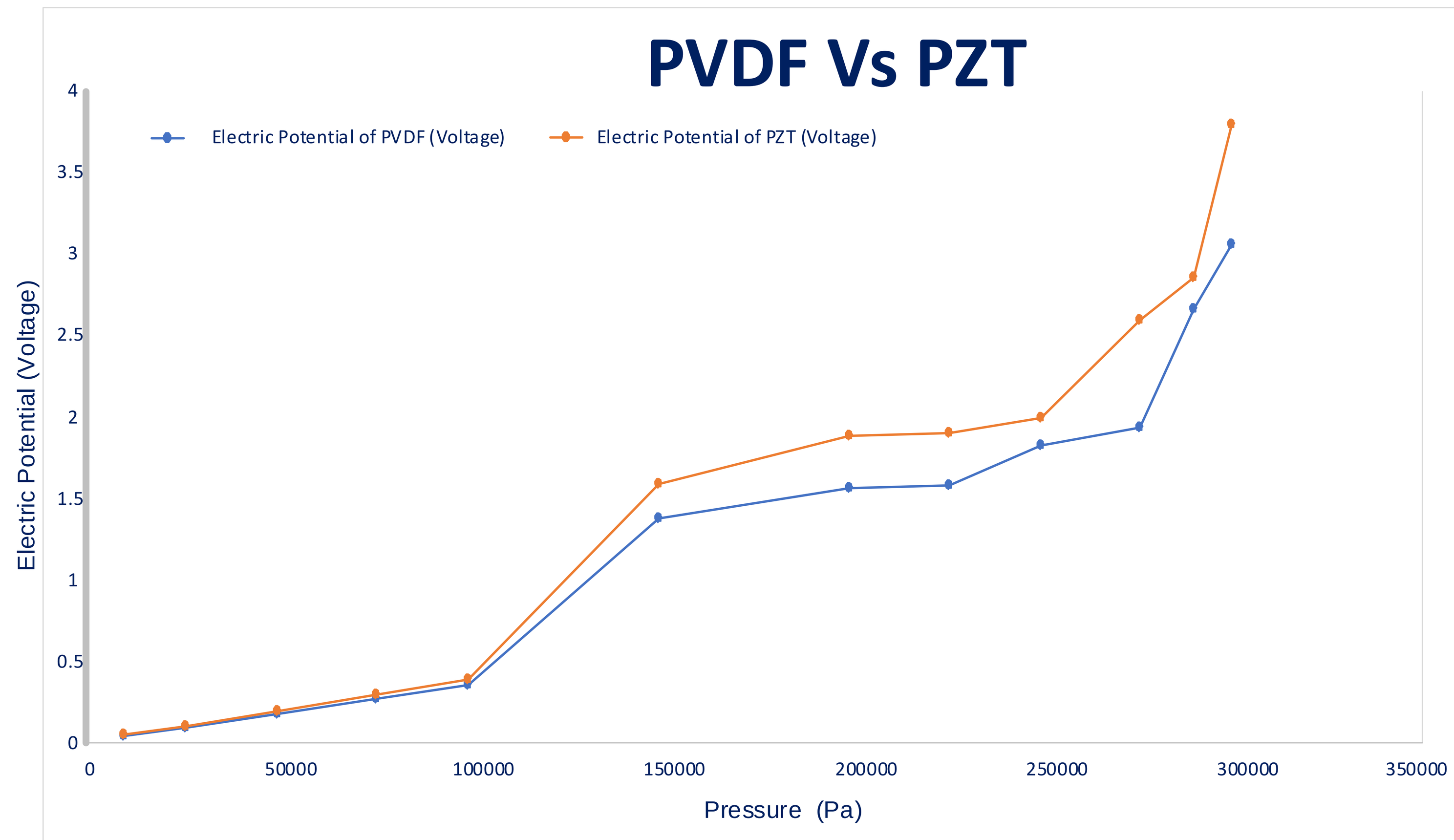

Fig.4. Electric Potential Vs Applied Pressure for PVDF and PZT

IV. CONCLUSION:

This study demonstrates the potential of PVDF and PZT piezoelectric materials for generating energy from human motion during everyday activities. PZT offers a much higher voltage output, while PVDF provides the flexibility needed for wearable and deformable devices. With appropriate load resistance and careful integration of both materials, these harvesters can serve as effective components in self-powered smart systems. Future work will focus on building physical prototypes, carrying out biomechanical evaluations, and incorporating dedicated power-management ICs.

References

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